tukan

Junior Statistician exercise

The Mexican nominal yield curve

The question

The Mexican federal government funds itself across maturities from 28 days to 30 years. Rates across those maturities describe a nominal yield curve, but the supplied monthly auction observations are not simultaneous quotes. The curve moves, and the nine rates do not move in lockstep.

We want you to answer three questions, in this order.

1. **Is there structure in how the nine maturities move together?**
2. **Can we forecast the curve** — one quarter and four quarters ahead?
3. **Does the structure in (1), if any, improve the forecasts in (2)?**

The questions build on one another. Three constraints apply to how you answer them.

- **Structural arguments should focus on the full curve.** Analysis of individual yields can support your conclusions.
- **Compare your forecasts against a naive baseline.** Tell us which baseline you chose and why. Compare methods on the same forecast origins and targets.
- **Define your forecast target and information set.** Explain what one and four quarters ahead mean for these monthly data, and what information was available at each forecast origin.

The data

One file, `banxico_auction_rates_monthly.csv`: monthly observations of the average annualised rate paid at auction, for nine instruments, from October 2006 to September 2026. Source is Banco de México, via Tukan's warehouse.

`DATE`	First day of the month the rate refers to
`INSTRUMENT`	CETEs at four maturities, MBONOs at five
`INDICATOR`	What the number measures
`VALUE_PCT`	The rate, in percent

The file preserves the available monthly observations. How you structure and model the data is up to you.

You may bring in other data, although this is not required. Macroeconomic and market data could help inform forecasts. If you use them, account for when each observation would have been available.

What to hand in

There are three deliverables:

1. **A log of your modelling decisions.** What you decided, why, what you rejected, and what you are unsure about. Include the things you tried that did not work. Any format and length is acceptable.
2. **A brief with your findings.** It should answer main questions around your submission such as data wrangling decisions, statistical modelling, and results comparison. It must be in PDF format and at most 5 pages.
3. **A presentation of what you found.** However you want to present it, assume a 20-minute slot.

We are not assessing typesetting, code style, file structure, or where you ran it. Use whatever tools you like. You should be able to explain and defend everything you submit.

How this is assessed

The assessment will have **two steps**.

The first step will be based on your submission only. We will review it and decide internally if we move into an interview. Feedback around your submission will be given regardless of your advancement.

The second step will be an interview around your presentation. The details will be given after passing the first step.

Practical

We expect this to take about 10 hours of work. If you find yourself on more than 20 hours of work, consider stopping: a narrow but well executed answer can be more compelling than a complex and convoluted approach with no explicit conclusion.

There is no single correct approach here. Several quite different ones lead somewhere defensible, and we are more **interested in how you got there than in which one you picked**.

These instructions go out on Friday, and we expect your answer by **Wednesday 23:59, Mexico City time**. Feel free to use either English or Spanish for the writeup; the interview will be in your preferred language of either of them.